Techminator – A Product Management Case Competition

You are SigmaCorp, a Technology Startup known for building revolutionary platforms. You are tasked with developing an innovative HR platform that consolidates multiple HR systems into a unified solution. The target audience for this platform includes Managers, HR personnel, senior leadership and employees from various organizations.

The platform's key objectives are to enable holistic analysis, eliminate bias in appraisals through intelligent features, and provide customized employee pathways for upskilling, while also handling basic HR systems like payroll. You are required to design a scalable platform that caters to both small and large organizations.

The intelligent features should encompass elements like AI-driven performance evaluations, sentiment analysis, or any other creative approaches to enhance the appraisal process. The platform should focus on the complete life cycle of the employee, right from talent identification and acquisition to Exit. Additionally, the platform should facilitate seamless data migration and integration from existing HR systems.

Participants must devise strategies that prioritize cost-effectiveness, ease of integration, and an exceptional user experience. They should consider the specific needs of different industries or domains while creating the pathways for employee upskilling.

Report Expectations (in any order)

- Problem Definition
- User research/ Data used.
- Persona and its Prioritization
- Proposing feasible features for the proposed product platform.
- Prioritizing the proposed solutions and their implementation
- Devising success metrics

Note: If the slide limit (max 10 slides + Appendix) condition is met, the participants can include any other information they believe is relevant to the case and their presentation.

<u>Judgment Criteria</u>

1. Creativity and feasibility of the solutions
2. The reasoning for prioritizing certain solutions.
3. Data backing/User research of the problem identification
4. Relevance of solutions finalized in the context of user benefits.
5. The type of success metrics chosen
6. Brownie points for including screenshots/wireframe designs in the report.

Note: Judges may include any other criteria they believe are important for the case. Judges' decisions will remain final

Problem Definition → Market Research → User Persona → User Journey → Proposed Features → Prioritisation → Success Metrics

PROBLEM STATEMENT

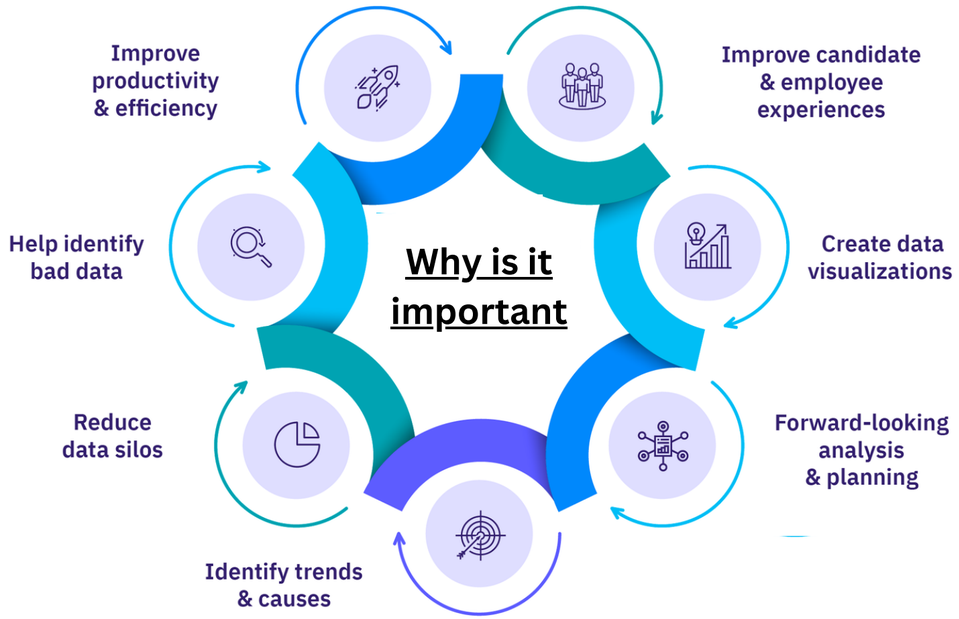
Developing an innovative HR platform that consolidates **multiple HR systems into a unified solution**

SCOPE

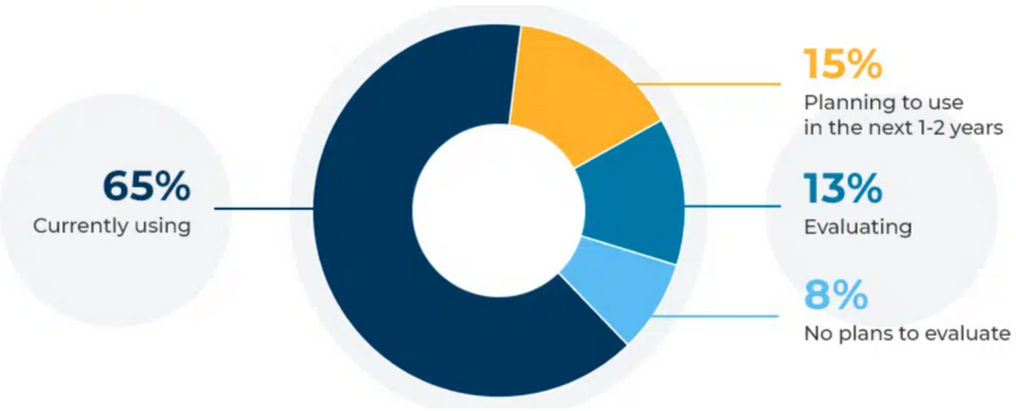
- Focus on the **complete life cycle of the employee**, right from talent identification and acquisition to Exit.
- **Scalable** - catering to both small and large organizations.
- Seamless **data migration** and **integration** from existing HR systems.
- **Customisable** to different industries.

IMPACT

- Increased **Efficiency** & **Saves Time** by reducing human dependency.
- More **data accuracy** by reducing errors and better reporting and visualisation.
- **Higher** level of employee **engagement** and **satisfaction** improving the overall experience.
- Availability of real-time data to make **informed**, strategic **decisions** aligned with business goals.
- **Cost Savings** by optimising processes and reducing manual efforts.
- Frees the HR teams to focus on strategic initiatives that add value to the organisation
- Can contribute towards building a **positive organisational culture** focused on growth and collaboration



Unified HR Solution Adoption in US (Small Companies)



CHALLENGES

- **Complex Implementation** to Integrate with existing HR Systems
- **Resistance to Change by** employees leading to slower adoption rates.
- Adapting the unified system to fit the organisation's specific needs may require **customisation.**
- Transitioning to a new system requires **training** for employees and managers.
- Initial investment for implementation, customisation, training, and ongoing maintenance can be **costly**.
- Integrating data from disparate systems can lead to **data quality issues.**
- The system needs to **scale effectively** to accommodate increased data volume and usage **without compromising performance.**

MARKET TRENDS

- The **rise of cloud-based unified HR systems**. A recent study by Gartner predicts that by 2023, 70% of all unified HR systems will be cloud-based.
- **Increased use of AI & ML** to automate tasks, improve decision-making, and provide insights into employee data.
- The increased **focus** on **employee experience.**
- The growing **importance** of **data analytics** to make better decisions about things like hiring, training, and compensation.
- The need for **security** and **compliance.**
- With the rise of remote and hybrid work models, there is an **increased offering of mobile applications**

Problem Definition → Market Research → User Persona → User Journey → Proposed Features → Prioritisation → Success Metrics

Competitor Feature Profile

Features →	Recruitment	Employee Onboarding	Time & Attendance	Payroll	Employee Engagement	Employee Upskilling	Performance Management	Rewards & Recognition	Employee Helpdesk	Travel & Expenses	Compensation planning	People Analytics	Remote work Suite
darwinbox	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
ORACLE HRMS	✓	✓	✓	✓	✓	✓	✓	✗	✓	✓	✓	✓	✗
SAP HR	✓	✗	✓	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗
monday	✓	✓	✓	✗	✓	✓	✓	✗	✗	✗	✗	✗	✗

Problem Definition → Market Research → User Persona → User Journey → Proposed Features → Prioritisation → Success Metrics

SHREYA

- **Demographics:**
  25, Female
- **Profession:**
  Software Engineer

- **Interests:**
  Travelling, Painting & Binge Watching
- **Goals:**
  To travel the world someday

- **Pain Points:**
1. She travels a lot by making impromptu plans but every time she has to **meet the manager in person to ask for leaves.**
2. She has to **coordinate with HR for attendance** every time she **works from home.**
- **Expectations:**
1. Better **leave management** and approval procedure.
2. **Seamless WFH experience**.
3. A **one-place help desk** to answer all her queries.

AVINASH

- **Demographics:**
  24, Male
- **Profession:**
  HR Executive Partner

- **Interests:**
  Reading, Gymming & Open Mic
- **Goals:**
  To learn accounting & to author a book

- **Pain Points:**
1. He is involved in talent recruitment but **cannot predict if a person will join after signing the offer letter** and is facing very low numbers recently.
2. He finds it **difficult to manage the expenses of multiple stakeholders simultaneously** on Excel.
- **Expectations:**
1. **Smooth** Employee **Recruitment cycle** to manage costs and resources
2. A **better way to handle travel and expenses** for multiple stakeholders.

YAJNA

- **Demographics:**
  28, Female
- **Profession:**
  Senior Manager - Product

- **Interests:**
  Dancing, Sketching & Hosting Events
- **Goals:**
  To launch her own startup

- **Pain Points:**
1. A lot of employees in her team are on notice period with better offers, she needs to make the **decision of who to keep & who to let go**
2. There have been complaints about some employees not performing well. She is unsure **how to measure their performance**.
- **Expectations:**
1. A **compensation planning suit** that helps her identify the valuable employee to retain
2. A day-to-day **performance management tool** for transparency and fair reward distribution

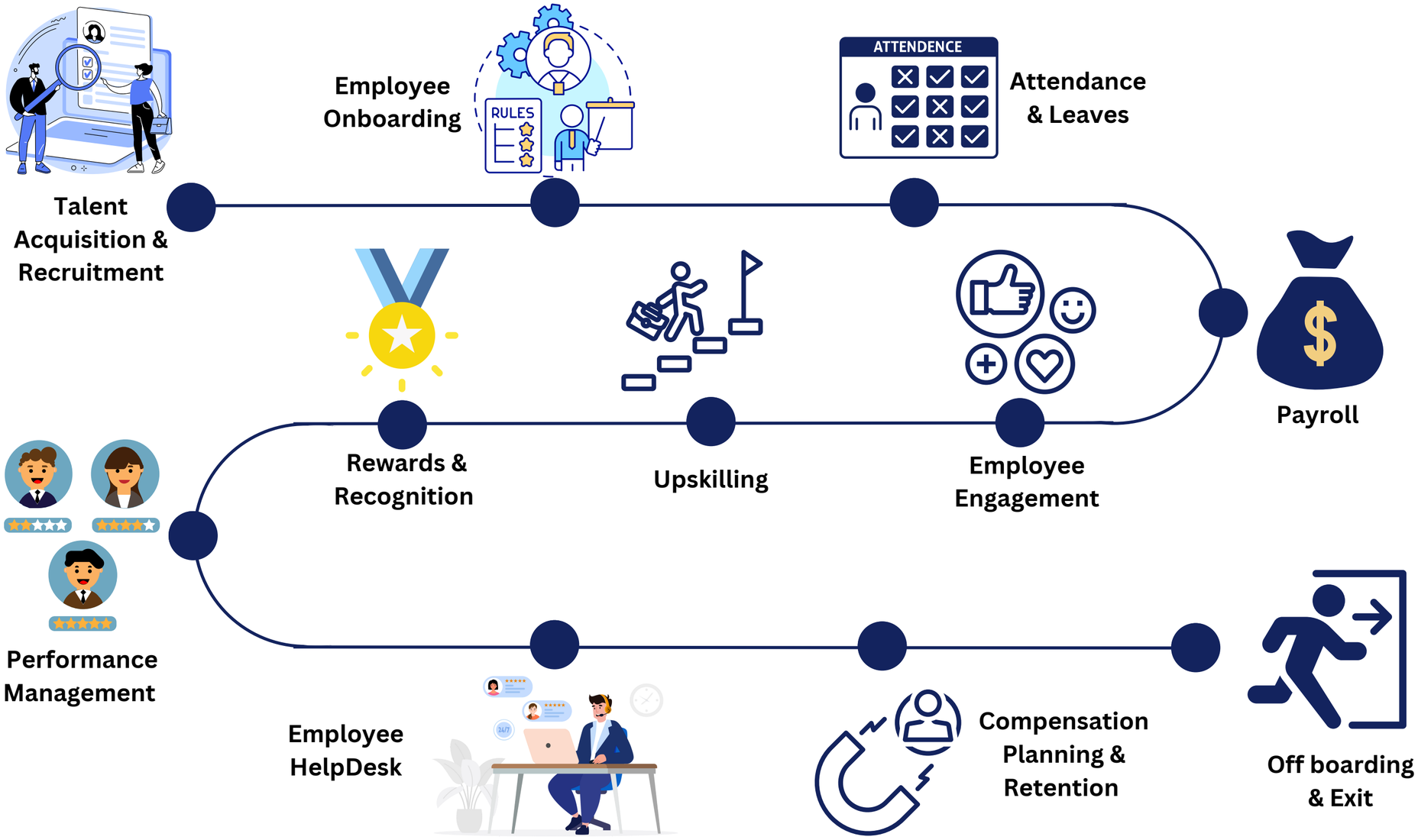
SAMAR

- **Demographics:**
  32, Male
- **Profession:**
  Vice President - Technology

- **Interests:**
  Trekking & Bike Riding
- **Goals:**
  To create impact in society through simple solutions

- **Pain Points:**
1. There is a significant **skill gap among employees** which is hindering innovation within the organisation.
2. **Employee disengagement** seems to be increasing and the reasons are not known.
- **Expectations:**
1. A **tool to** help employees identify their skills, their skill gap and help them **upskill**
2. An **employee analytical platform** to gain insights about what factors contribute to employee engagement and satisfaction and to take necessary steps accordingly

Problem Definition → Market Research → User Persona → User Journey → Proposed Features → Prioritisation → Success Metrics



Step	Persona Involvement
Talent Acquisition & Recruitment	Employee, HR
Employee Onboarding	Employee, HR, Manager, Senior Leadership
Attendance & Leaves	Employee, HR, Manager
Payroll	Employee, HR
Employee Engagement	Employee, HR, Manager, Senior Leadership
UpSkilling	Employee, HR, Manager, Senior Leadership
Rewards & Recognition	Employee, HR, Manager
Performance Management	Employee, HR, Manager, Senior Leadership
Employee HelpDesk	Employee
Compensation Planning & Retention	Employee, HR, Manager, Senior Leadership
Off boarding & Exit	Employee, HR

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MVP FEATURES

1 Recruitment/Onboarding

- Manage job postings across multiple platforms.
- Screen and rank applicants.
- Streamline the onboarding process with checklists, document management, and orientation scheduling.

2 Performance Management

- Set and monitor employee goals.
- Conduct performance reviews.
- Manage feedback and coaching.

3 Core HR & Payroll

- Personnel Administration
- Organizational Management
- Payroll Processing and Taxation
- Time and Attendance Management
- Benefits Administration
- Global HR Compliance

4 Time & Attendance

- Track employee working hours, absences, and leave requests.
- Plan shifts and rotations.
- Ensure compliance with labor laws and regulations.

5 Employee Upskilling

- Curate content and create learning paths.
- Offer eLearning, classroom training, and blended learning experiences.
- Track and report on training completions and compliance.

6 Employee Engagement

- Feedback and Surveys: Gather feedback from employees through surveys.
- Well-being: Monitor and promote employee well-being and health.
- People Analytics: Advanced analytics and reporting tools for HR data.

7 Holistic Analysis Dashboard

Unified View

- **Consolidates data** from multiple sources.
- Offers a **comprehensive view** of organizational performance.
- **Simplifies access** for decision-makers, eliminating the need to switch between systems.

Real Time Monitoring

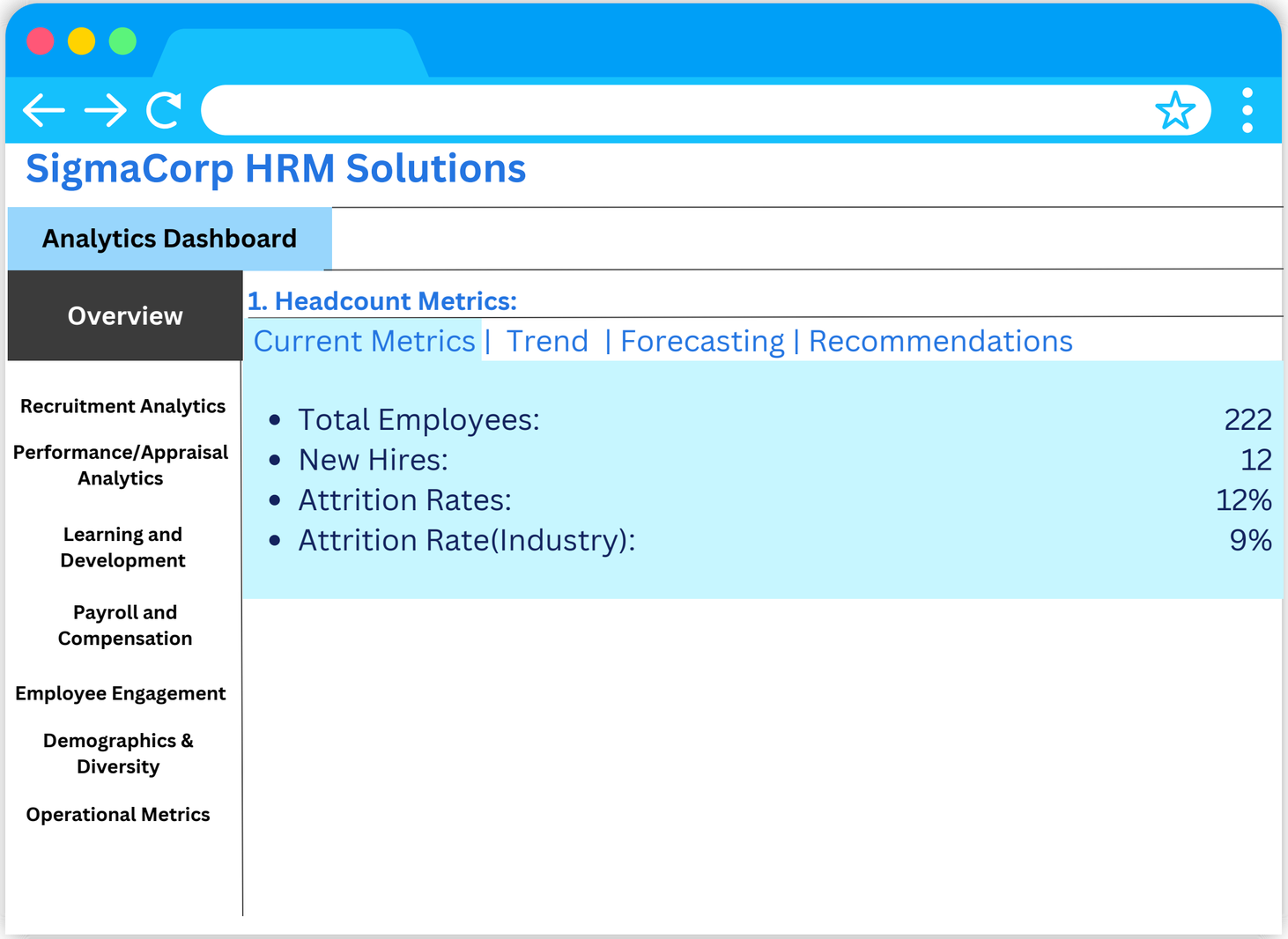
- Offers **updates in real-time.**
- Enables **immediate insights** into current data.
- Facilitates swift responses to new situations or trends.

Recommendation and Alert System Using AI

- Utilizes advanced algorithms to **analyze complex data** sets.
- Provides **personalized recommendations** based on user behavior or preferences.
- **Proactively detects anomalies** or outliers and sends alerts.

Trend Identification

- Analyzes historical and current data to spot patterns.
- Helps in **predicting future behaviors** or outcomes.
- Enables proactive strategies based on observed trends.

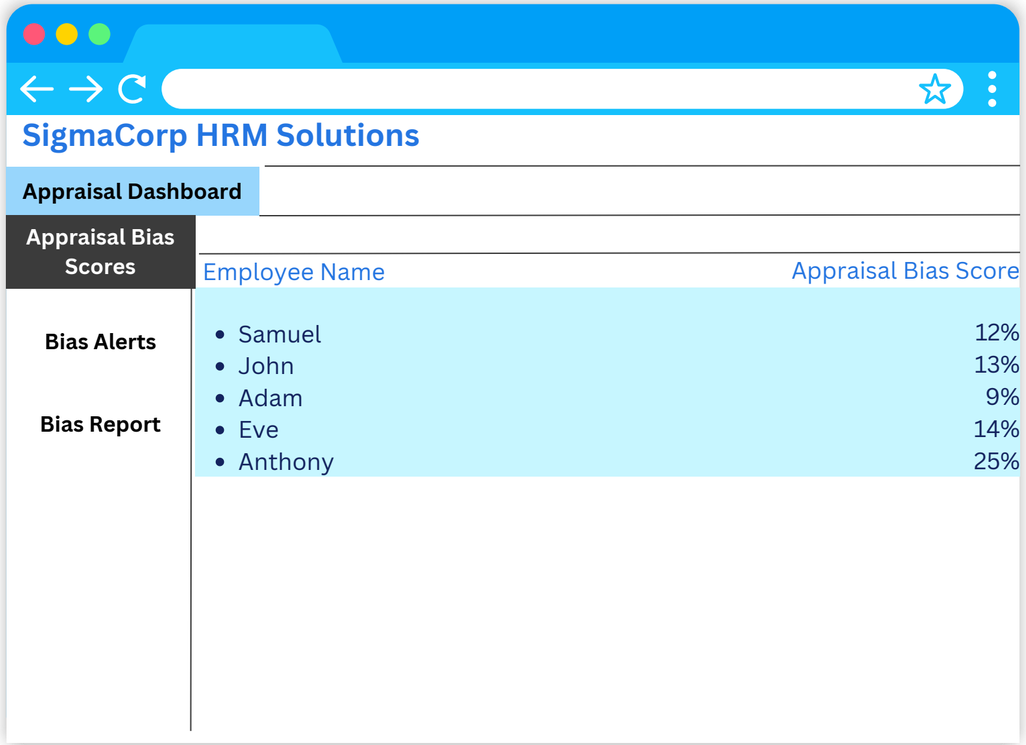


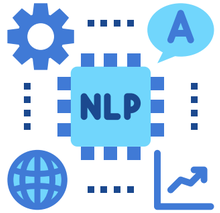
8 Appraisal Bias Tracking Tools

Appraisal Bias Score

Using some Bias Detection Algorithms:

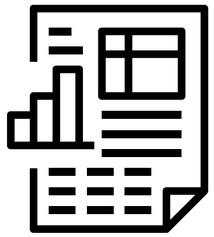
- Generate **Bias Score** for each employee
- More bias score represents more bad effect on employee appraisal due to bias.
- **Function:** Analyzes performance evaluation data to identify patterns that may indicate bias. For example, if one group consistently receives lower scores than others despite similar output or achievements, the algorithm could flag this as potential bias.
- **Benefits**: Provides an objective, data-driven check on human evaluations; offers actionable insights to HR teams.





Review Alerts

- **Function**: Analyzes written feedback to detect potentially biased language or phrases using **NLP**
- **Benefits:** Helps reviewers become more aware of their language choices and promotes more objective, fact-based feedback.



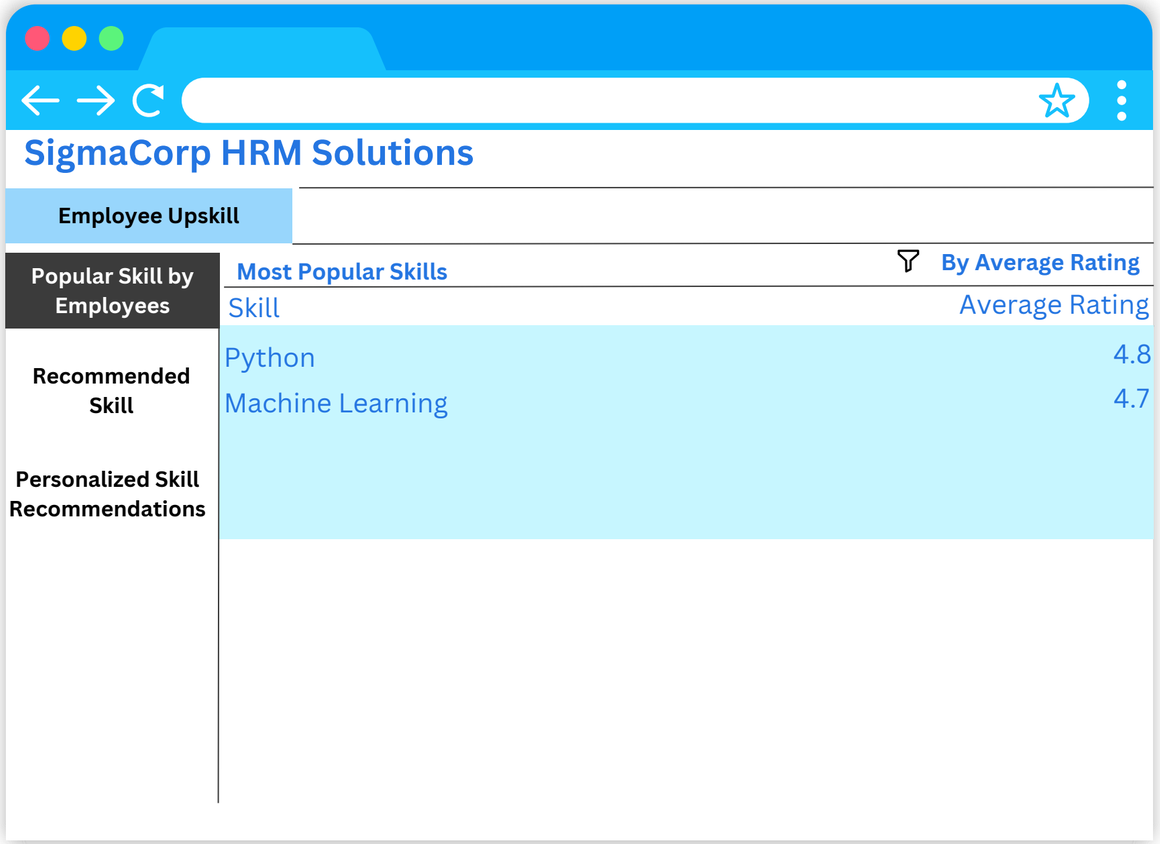
Bias Reports

- **Function:** Allows HR to evaluate team/organization appraisal bias report as a whole
- **Benefits:** Give a viewpoint to the HR on the inclusivity, employee wellness and satisfaction scenario in the team/organization

Customizable Alerts

- **Function:** Allows HR teams to set specific triggers or thresholds for bias alerts. For example, if a particular reviewer consistently gives lower scores to a certain demographic, the system could flag this behavior.
- **Benefits:** Offers flexibility in monitoring and addressing potential biases, tailored to the organization's unique needs.

9 Learning/Training Path Recommendation

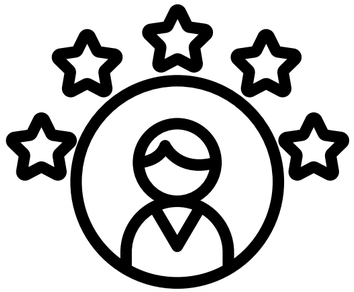


Popular Skill among Peer

- **Function:** AI algorithms analyze discussions, feedback, and ratings from peers to recommend popular or highly-rated training modules.
- **Benefits:** Incorporates peer insights into learning recommendations, fostering a sense of community and shared learning.

Personalized Path Recommendation

- **Function:** AI algorithms analyze an individual's role, past training, performance metrics, and career goals to curate a personalized list of courses or training modules.
- **Benefits:** Ensures that employees receive training that is most relevant to their current roles and future aspirations, increasing the effectiveness of training initiatives.



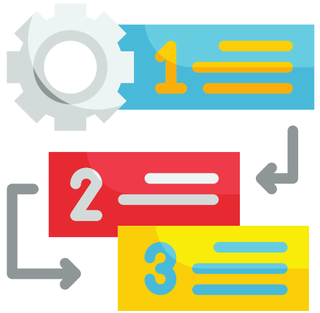
Recommended Skill to achieve organizational goal

- **Function**: Analyzes written feedback to detect potentially biased language or phrases using **NLP**
- **Benefits:** Helps reviewers become more aware of their language choices and promotes more objective, fact-based feedback.

Prioritization (Using RICE framework)

Phase 1 (MVP features)

Feature	Reach	Impact	Confidence	Effort	RICE Score
Recruitment/Onboarding	8	3	90%	8	2.70
Performance Management	7	3	85%	7	2.55
Core HR & Payroll	9	3	95%	9	2.85
Time & Attendance	8	2	90%	6	2.40
Employee Upskilling	6	2.5	80%	7	1.71
Employee Engagement	6	2	80%	6	1.60



Phase 2 (Advanced features)

Feature	Reach	Impact	Confidence	Effort	RICE Score
Holistic Analysis Dashboard	9	2.5	80%	8	2.25
Appraisal Bias score implementation	7	3	85%	7	2.55
Bias reports for teams and organization	8	2.5	80%	6	2.67
AI Customizable Alerts for bias in appraisal	7	2.5	85%	7	2.13
Appraisal Review alerts using NLP	7	2.5	80%	8	1.75
Learning path by popular skill among peers	6	2	80%	6	1.60
Personalized path Recommendation using AI	6	2.5	80%	7	1.71
Skill recommendation to achieve org goal	8	3	85%	7	2.91

Problem Definition → Market Research → User Persona → User Journey → Proposed Features → Prioritisation → Success Metrics

Success Metrics (Top 2 recommended feature)

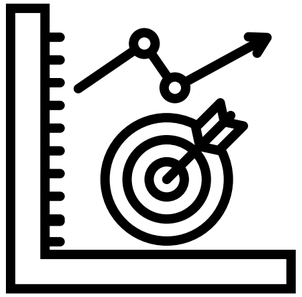
Phase 1 (MVP features)

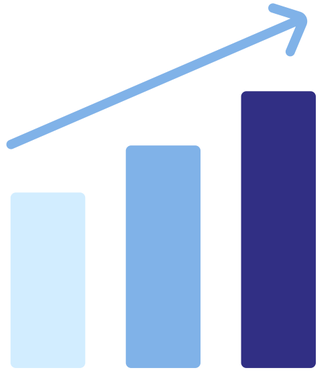
1. Core HR & Payroll:

- **Data Accuracy Rate:** Percentage of payroll and HR records that are error-free. A high accuracy rate indicates that the system is reliable, reducing the likelihood of complaints or disputes related to payroll or HR data.
- **Time-to-Payroll:** The average time taken from the end of a pay period to the disbursement of salaries. A reduction in this time indicates efficiency improvements in the payroll process.

2. Recruitment/Onboarding:

- **Time-to-Hire:** The average number of days from the posting of a job to the successful onboarding of a candidate. A reduction in this metric can indicate that the recruitment module is streamlining the hiring process.
- **New Hire Retention Rate:** The percentage of new hires who remain with the company after a certain period (e.g., 90 days, 6 months). A high retention rate suggests that the onboarding process is effective, and new hires are well-integrated.





Phase 2 (Advanced features)

1. Skill Recommendation to Achieve Organization Goal:

- **Skill Application Rate:** The percentage of employees who apply the skills they've learned within a specified timeframe after completing a recommended training. A high rate indicates that the training was relevant and valuable.
- **Performance Improvement Post-Training:** Measure the average performance score improvement of employees post-training compared to their pre-training scores. A notable increase suggests that the skill recommendations are positively impacting performance.

2**. Bias Reports for Teams and Organization:**

- **Bias Incident Reduction:** Track the number of reported bias incidents or complaints before and after the implementation of bias reports. A reduction indicates that the system is effectively highlighting and addressing potential bias areas.
- **Diversity in Promotions**: Monitor the diversity of employees receiving promotions or positive performance reviews. A balanced representation post-implementation suggests that the bias reporting system is promoting fairness in appraisals.